# OpenADMET CYP Challenge — Inhibition Potency (Non-TDI) Track Model Description Report

**Submitted by:** Cheminfosolutions Ltd · **leaderboard alias:** Cheminfo
**Predicted property:** pIC50 of direct inhibition for CYP1A2, CYP2C9, CYP2D6 and CYP3A4; blind set 750 compounds
**Metric:** ST-RAE (soft-threshold relative absolute error), macro-averaged over the four isoforms
**Training data:** challenge-distributed data only; no external corpus contributes to any submitted column

## 1. Models

| Arm | Model / architecture | Molecular representation | Reference | In final file |
|---|---|---|---|---|
| **ChemProp 8-task** | D-MPNN trunk, 4 scored pIC50 heads (weight 1.0) + 4 auxiliary single-shot log2fc heads (weight 0.2); hidden 400, depth 4, 3-layer FFN, dropout 0.15, batch 128; 15-member ensemble = 5 seeds × 3 split methods (random, stratified, scaffold); band loss α = 0.7; BoundedMSE censoring; predictions clipped to [1.5, 9.0] | Learned bond-message graph embedding + RDKit 2D descriptor block concatenated at the FFN | Heid et al., *J. Chem. Inf. Model.* **2024**, 64, 9–17; Yang et al., *ibid.* **2019**, 59, 3370–3388 | Yes — CYP2C9, CYP2D6 |
| **Primary + proxy blend** | 8-head model on real labels paired with a proxy model trained only on screened-but-never- fitted compounds, via a fold-internal single-dose → pIC50 calibration; per-isoform blend weight chosen out of fold; censoring at 4.305 and inverse credible-interval weighting | as above | as above | Yes — CYP3A4 |
| **Multi-member consensus** | 23 members, equal weight, each z-scored within itself; members deduplicated by ranking at ρ ≥ 0.999; affinely placed | member-dependent | internal | Yes — CYP1A2 |
| LightGBM | Gradient-boosted trees; member of the CYP2C9 order-only rank perturbation | RDKit descriptors + Morgan / FCFP counts + MACCS keys | Ke et al., *NeurIPS* **2017**, 3146–3154 | Order only |
| Published-protocol grid | SVM / RF / kNN / ANN over PaDEL fingerprints, per the published CYP protocol, retrained on the challenge labels | PaDEL fingerprints (AP2D, CDK, CDKExt, EState, GraphFP, MACCS, KRFP, PubChemFP, SubFP) | Xu, Lu, Wu, Gui, Liu, Tang & Li, *Mol. Pharmaceutics* **2023**, 20, 194–205 | Order only |
| MLP | Feed-forward network on the descriptor block; member of the CYP2C9 perturbation | RDKit descriptors | scikit-learn (Pedregosa et al., *JMLR* **2011**, 12, 2825–2830) | Order only |
| ChemProp, CheMeleon trunk | Pretrained foundation D-MPNN, hidden 2048, depth 6, 8.7 M parameters, as an alternative trunk | as ChemProp | CheMeleon foundation weights distributed with ChemProp v2 | No |
| XGBoost QSAR | In-house boosted-tree QSAR arms built as CYP3A4 candidates | RDKit PhysChem + RDK / AtomPair / Torsion / Morgan fingerprints | Chen & Guestrin, *KDD* **2016**, 785–794 | No |

**Descriptors.** The neural arms concatenate the RDKit 2D physicochemical block (~200 descriptors) to the learned graph embedding at the feed-forward layer; the tree and kernel arms use that block plus count-based Morgan and FCFP fingerprints and MACCS keys; the published-protocol arm uses PaDEL fingerprints as published. Non-finite columns and columns exceeding 1E10 are dropped before fitting. Software: ChemProp 2.2.3 with PyTorch Lightning, RDKit (Landrum, rdkit.org), PaDEL (Yap, *J. Comput. Chem.* **2011**, 32, 1466–1474), LightGBM, scikit-learn, Schrödinger LigPrep; training on an NVIDIA T4.

**Loss and selection.** ST-RAE charges nothing inside a compound's credible interval, so the training loss carries a matching band term per labelled cell, in scaled target space: `L = α·[relu(pred − (y+hi)) + relu((y−lo) − pred)] + (1−α)·(pred − y)²`, with α = 0.7 and `hi`, `lo` that cell's own asymmetric half-widths. Arms were accepted per isoform on a fixed gate of mean error, 90th-percentile error and ranking against two baselines.

## 2. Results

| Isoform | ST-RAE | Mean | SD | Submitted configuration, post-processing in order |
|---|---|---|---|---|
| CYP1A2 | **0.5090** | 4.4746 | 1.0779 | 23-member z-score consensus → placed on recovered moments (4.32 / 1.39) → floor 3.1194 |

| Isoform | ST-RAE | Mean | SD | Submitted configuration, post-processing in order |
|---|---|---|---|---|
| CYP2C9 | **0.4265** | 4.8805 | 0.7977 | 8-task band-loss ChemProp, tier-S pseudo-labels → floor 3.2130 → QSAR rank perturbation w = 0.10, quantile-mapped (order only) |
| CYP2D6 | **0.5072** | 3.6979 | 0.8386 | 8-task band-loss ChemProp, tier S (2,424 pseudo cells) → floor 3.1478 (440 rows) |
| CYP3A4 | **0.5057** | 4.7967 | 0.9612 | Primary + proxy blend → floor 3.1079 (91 rows, pile held) → floor-held attenuation-matched narrowing, scale 0.9145 on the 659 rows above the floor |
| **Macro-average** | **0.4871** | — | — | 750 rows, zero sub-floor predictions on any isoform; validated with the challenge's own activity validator |

Board read 2026-11-02. Prediction floors are each isoform's minimum observed credible-interval upper bound; blind-set moments used for affine placement were recovered from the public leaderboard, with the scale target $\rho \cdot \sigma_y$.

## 3. Data, ligand preparation and labels

| Dataset | Role | Size |
|---|---|---|
| `TRAIN_inhibition` | Dose–response pIC50, primary labels (CYP1A2 1,412 · CYP2C9 1,285 · CYP2D6 1,493 · CYP3A4 2,335) | 4,905 compounds |
| `single-concentration-TRAIN` | Single-shot screen: auxiliary heads and pseudo-labelling | 4,376 × 4 isoforms |
| `TEST-BLINDED` | Blind prediction set | 750 compounds |

**Ligand preparation**, applied identically to training and blind sets; the six RDKit corrections run in the order shown, through a single driver.

| # | Step | Detail |
|---|---|---|
| 0 | Schrödinger LigPrep | Preparation at pH 7.4 |
| 1 | Sulfonamide N | R-$SO_2$-[$N^-$]-R′ → R-$SO_2$-NH-R′ |
| 2 | Pyridinium N | [$nH^+$] → n; quaternary [$n^+$] left unchanged |
| 3 | Morpholine N | [$NH^+$] → N ($pK_a$ ≈ 6.5, not dominant at pH 7.4) |
| 4 | Iminium N | Amidine / guanidine / iminium C=[$NH^+$] → C=N, $sp^2$ nitrogen only |
| 5 | Amide | `N=C([O-])` → `N-C(=O)`, bond orders corrected |
| 6 | Phenol | `c[O-]` → `c-OH` |
| 7 | Artefact fallback | Azolium, iminol, iminium and imine tautomers revert to the raw input SMILES. Net effect: 252 of 4,905 training rows and 52 of 750 test rows change; charged molecules 941 → 718 (train) and 128 → 80 (test) |

**Label interpretation.**

| | |
|---|---|
| **Censoring** | Dose–response readings at or below the top screening concentration (pIC50 4.305) establish an upper bound only, and are left-censored under BoundedMSE |
| **Missingness** | About one third of screened compounds received a curve; for CYP2D6 the curves were run preferentially on apparent hits, so all 1,493 CYP2D6 labels are inhibitors while 2,883 inactive compounds carry an empty cell |
| **Pseudo-labels (tier S)** | The curve-fitter's fallback value is a software floor, agreeing across isoforms to within 0.04 log units (1A2 2.168 · 2C9 2.333 · 2D6 2.168 · 3A4 2.206). Each isoform's own floor is assigned to every compound in the clearly-inactive zone of the screen: 2,687 cells added, 2,424 of them CYP2D6 non-inhibitors, contamination 2.3%, mean error 0.168. CYP2D6 labels 1,493 → 3,917 |
| **Pseudo-label treatment** | Entered as point estimates, not censored bounds: censoring is enabled for real measurements and disabled for pseudo rows, with a hard abort if the CYP2D6 prediction standard deviation falls below 0.5 |
| **External data** | A 30,980-row union corpus was assembled and not used: ChEMBL CYP potency carries a +0.45–0.65 log offset and a hard floor at 4.00 against this assay. Where external data was used in development it was carried on separate heads, never merged into a scored target |